Negative-Core Sample Knowledge Distillation for Oriented Object Detection in Remote Sensing Image

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***Abstract*— Knowledge distillation (KD) has been one of the most effective methods for enhancing the performance of lightweight detectors, crucial for remote sensing edge intelligence models. However, many mainstream distillation methods that are centered around the paradigm of distilling positive samples show weak exploitation of the student's potential. This arises due to these methods overlooking the core teacher–student difference in remote sensing scenarios with vast and object-similar backgrounds. In this article, from the point of distillation sample and knowledge hierarchy, we design a negative-core sample knowledge distillation (NSD) method for improving the performance of the lightweight object detection model. Specifically, a negative-core sample (NCS) is innovatively employed to transfer effective background discrimination knowledge for bridging the core difference. KD for NCS across four levels—pixel, logit, box, and angle—are customized to fully leverage the teacher's insights. Category direction estimation (CE) is incorporated into the angle KD to convey NCS-oriented knowledge more effectively. Extensive experiments conducted on multiple remote sensing datasets achieve state-of-the-art (SOTA) performance, demonstrating the effectiveness of the proposed NSD. Codes are available at https://github.com/Changan00/NSD.**

***Index Terms*— Deep learning (DL), knowledge distillation (KD), oriented object detection, remote sensing imagery.**

I. Introduction

THE rapid evolution of deep convolutional neural networks (CNNs) has significantly enhanced the application of object detection within the realm of remote sensing. This encompasses a range of activities from satellite orbit processing and mission planning to unmanned aerial vehicle (UAV) tracking and navigation. However, object detection in remote sensing presents unique challenges. Unlike objects found in natural scenes, those in remote sensing imagery often blend with their background, displaying similar textures, colors, and shapes, and are typically found in dense clusters with arbitrary orientations. To effectively distinguish these objects, detectors typically rely on large neural network architectures capable of generating discriminative features and incorporating additional directional information to maintain detection accuracy. Nonetheless, this requirement for complex models leads to substantial computational and storage demands, posing a significant hurdle for deployment on end devices. In response to this challenge, the technique of knowledge distillation (KD), as introduced by Hinton et al. [1], has been adopted to mitigate these issues.

In recent years, KD has gained widespread application across detection tasks within the domain of natural scene analysis. As a strategy for model compression, KD focuses on enhancing the performance of a less complex student model by leveraging insights from a more sophisticated teacher model. The essence of KD in the context of detection revolves around the nature of the knowledge being transferred, which can broadly be categorized into two types: feature-based distillation and prediction-based distillation. Feature-based distillation [2], [3], [4], [5], [6], [7], [8], [9] focuses on improving the alignment of deep features between the teacher and student models by minimizing the L1 loss between their respective features. The majority of existing approaches concentrate on distilling the features from positive samples. On the other hand, prediction-based distillation is derived from methods specific to classification tasks. This technique transfers semantic knowledge from the logit of positive samples in the classification branch, along with localization knowledge from bounding boxes in the regression branch of these samples [10], [11], [12], [13], [14].

The above studies have extensively promoted the development of the KD field. Nevertheless, the approaches of distilling positive samples overlook the remote sensing characteristics with large and object-similar backgrounds. The following challenges exist in remote sensing KD.

1) The ability to identify positive samples between the teacher and student models exhibits minimal differences. As shown in Fig. 1, the recall rates for certain categories are nearly identical between the student and teacher models, indicating that both models can recognize a similar proportion of objects within the positive

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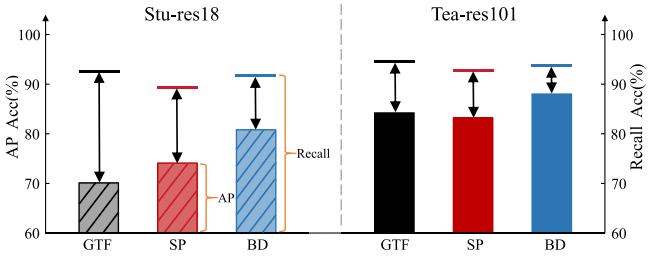


Fig. 1. Comparison of the partial classification of student and teacher on the DOTA dataset based on the RetinaNet detector, where Tea-re101 represents teachers and Stu-res18 represents students. GTF, SP, and BD are several typical categories in the DOTA dataset. The two-way arrow represents the gap between AP and Recall in each category. The left vertical axis represents the AP accuracy, corresponding to the long bars, and the right vertical axis represents the Recall accuracy, corresponding to the horizontal lines above the long bars.

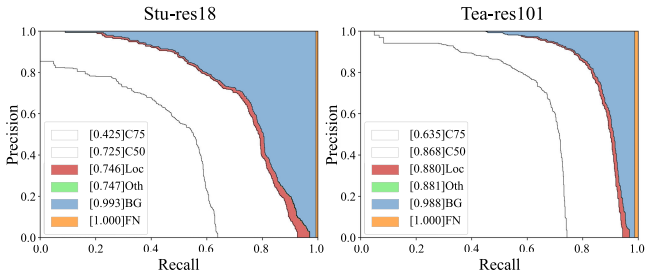


Fig. 2. Precision–recall map of GTF. Tea-re101 represents teachers and Stu-res18 represents students. C75: correct category (IoU $>$ 0.75). C50: correct category (IoU $>$ 0.5). Loc: correct category but misaligned box (0.1 $<$ IoU $<$ 0.5). Oth: wrong category (IoU $>$ 0.1). BG: background false positives (IoU $<$ 0.1). FN: false negatives (remaining errors).

samples. However, a noticeable gap exists in average precision (AP), with the student model significantly lagging behind the teacher, indicating that the student model incorrectly detects more objects.

2) The background and the object exhibit similarities in texture and shape, which leads the more compact student model to incorrectly classify numerous backgrounds as objects. As shown in Fig. 2, the student model, in comparison to the teacher, demonstrates a higher incidence of false positives. More intuitively, as shown in Fig. 3, the disparities between the student and teacher models in the object regions are minimal, whereas the differences in the background are notably more pronounced.

With the above analysis, it is crucial to improve students' background discrimination ability to reduce the performance differences between teachers and students. A factor in the weak generalization of the existing KD methods in the remote sensing scene is that those methods focus on distilling knowledge from positive samples, and the transference of background discrimination knowledge is ignored.

Exactly as "There are also pearls in the sand grains," even in seemingly unremarkable entities like sand, valuable insights can be found. Similarly, negative samples can effectively supplement the learning of the student model. To harness this potential without introducing noise inherent in negative samples, we meticulously identify negative-core samples (NCSs) by adopting methods tailored to various structures. Based on the NCS, we propose a negative-core sample knowledge distillation (NSD) method and employ the NCS in knowledge across four levels—pixel, logit, box, and angle—to transfer effective background discrimination knowledge.

The NSD method consists of two components: negative-core sample feature assimilation (NFA) and negative-core sample anti-false alarm prediction (NAP). NFA meticulously selects NCS by adopting methods tailored to various structures. Meanwhile, it uses teacher NCS features for effective pixel-level feature distillation. NAP decouples the prediction knowledge into three levels: logit, box, and angle for more adequate distillation. NCS ensures the distillation of logit knowledge remains impervious to the influence of remote sensing characteristics. Box knowledge aids the student in achieving more precise localization. During the distillation of angular knowledge, given that a stepwise escalation in sample loss [15], we devise a novel category direction estimation (CE) approach that enables the transfer of discretized angle knowledge by minimizing the Kullback–Leibler (KL) divergence between the angle category logits of teacher and student predictions.

By transferring effective background discrimination knowledge, our NSD can reduce false alarm detections of objects in remote sensing images and achieve state-of-the-art (SOTA) results. The contributions can be summarized as follows.

1) An NSD framework is proposed for improving the performance of a lightweight object detection model. It innovatively adopts NCS as the carrier of knowledge, which enables the student model to acquire effective background discrimination knowledge from the teacher model.
2) A four levels of customized distillation for NCS are devised, encompassing pixel, logit, box, and angle. The knowledge related to features and predictions is concurrently fully leveraged.
3) A novel CE is developed, which transforms continuous angle regression problems into discrete angle distribution fitting problems and is more conductive to the transfer of NCS directional knowledge.

We conduct experiments on the DOTA [16], DIOR [17], and HRSC [18] datasets. Comprehensive experiments and comparisons with the SOTA demonstrate the effectiveness of our method. The remainder of this article is organized as follows. Section II provides a brief review of related work, Section III elaborates on the design and principles of the NSD method, and extensive experiments are presented in Section IV. Finally, the conclusion is outlined in Section V.

II. Related Works

A. Oriented Object Detection

Object detection is a fundamental task in computer vision with wide-ranging applications. Over the past decade, significant progress has been made in the development of object detection techniques. Numerous methods have emerged for object detection in natural scenes using a horizontal bounding box (HBB) [19], [20], [21]. In recent years, object detection in remote sensing scenes has gained significant attention due to its wide range of applications. However, remote sensing

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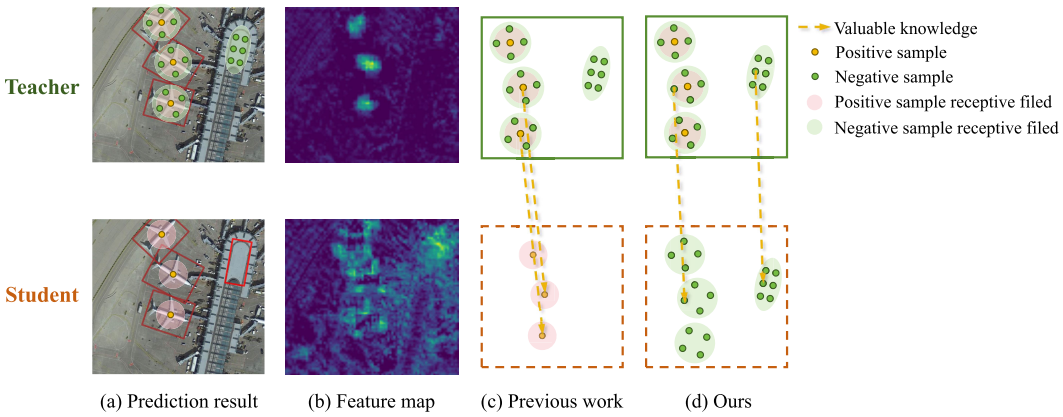


Fig. 3. Prediction result and feature of both the teacher and the student, along with a brief description of the distillation process. (a) Three dark red bounding boxes on the left side of the image represent planes detected by both the teacher and the student, while the bright red bounding box on the right side represents a false alarm GTF detected by the student. (b) Features extracted by the student focus on many irrelevant backgrounds, leading to misidentifying the background as the object. (c) Previous methods transfer knowledge of positive samples closely related to the object, but this cannot help the student in correctly recognizing the background. (d) Our approach transfers knowledge of negative samples, even though these negative samples do not closely match the object (IoU less than the threshold). However, (a) demonstrates that negative samples can still capture the knowledge of the object, which helps the student in correctly recognizing the background while further improving object recognition.

images are often captured from a top-down perspective, resulting in a large number of objects with arbitrary arrangements and different orientations. These objects are typically densely distributed and exhibit large aspect ratios, posing significant challenges for object detection tasks. Therefore, the adoption of oriented bounding box (OBB) detection, represented as (x, y, w, h, θ), has become a popular solution among many researchers to address these challenges.

Some works initially concentrate on the incorporation of angle within the regression process of the detection head [22], [23], [24], [40]. Rotated-RetinaNet (R-RetinaNet) [20] and R-Faster R-CNN [22] directly introduce the OBB regression module to handle oriented objects, which both provide a generic framework for subsequent methods. RRPN [23] adapts the oriented proposal to a two-stage network based on R-Faster Rcnn, and SCRDet [24] introduces a rotation-aware feature pyramid network (RoI-RPN) based on this for generating candidate regions. Oriented R-CNN [40] further enhances the detection capability of objects with diverse orientations by introducing an orientation-aware loss. However, oriented proposals are denser compared to horizontal proposals, resulting in speed degradation and imbalance issues. Some methods alleviate the aforementioned problems by improving horizontal proposals [25], [26], for example, region of interest (RoI) Transformer [25] converts horizontal RoIs to rotated RoIs, thus avoiding the direct generation of densely redundant oriented proposal boxes. S2A-Net [26] proposes the FAM module based on the single-stage R-RetinaNet to generate high-quality oriented anchor boxes.

The features for oriented object detection need to adapt to objects in various orientations, and some works have researched feature extraction [27], [28]. For instance, R3Det [27] introduces a feature refinement mechanism (FRM) based on RetinaNet, which effectively captures detailed information of oriented objects by inserting rotation-aware feature refinement modules into the feature pyramid network (FPN). Redet [28] establishes a rotation-equivariant feature representation in the network through the use of rotation convolution and rotation pooling operations, introducing rotation equivariance.

There have been works addressing the optimization of angle loss in the bounding box regression process [15], [29], [30]. CSL [15] introduces smooth labels to represent the bounding boxes of arbitrarily oriented objects. KLD [29] utilizes KL divergence to quantify the discrepancy between the detection results and ground-truth (GT) bounding boxes and minimizes this discrepancy to learn highly accurate rotated object bounding boxes. GWD [30] employs Gaussian distributions to model the position and size of the object bounding boxes and measures the discrepancy between the predicted and true Gaussian distributions by computing Gaussian Wasserstein distance. These oriented detection methods have demonstrated excellent accuracy in the field of oriented object detection. However, they often require larger backbone networks to extract high-quality features, which introduces a significant computational burden and makes it challenging to achieve fast response in resource-constrained scenarios such as edge devices.

B. Knowledge Distillation

KD is an excellent approach to address the aforementioned issues, which is a widely applied model compression technique. KD, proposed by Hinton et al. [1], aims to transfer knowledge from a teacher model to a student model. Based on the form of transferred knowledge, current KD methods can generally be classified into two categories: feature-based distillation and prediction-based distillation.

Feature-based distillation aims to transfer the intermediate layer feature knowledge from the teacher model to the student model. FitNets [2] demonstrates that the intermediate layer features of the teacher can serve as guidance for the student in classification tasks. Subsequently, numerous works have employed feature-based distillation methods for compressing object detectors [3], [4], [5], [6], [7], [8], [9]. Chen et al. [3] indiscriminately distilled the features from the neck of the detector, pioneering the application of KD in object detection. Li et al. [4] argued that sampled RoI regions contain valuable

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object-related knowledge. Wang et al. [5] distilled fine-grained features of foreground object regions, while Zhang et al. [7] performed feature distillation on foreground regions at different layers and scales. However, these methods lack the utilization of background region features. Yang et al. [8] recognized the role of background region features as a complement to foreground region features. Nevertheless, they pointed out that the coupling of foreground and background regions may have negative effects. These methods have shown promising results in natural scenes with well-arranged objects but are not suitable for remote sensing scenes with cluttered and densely distributed objects. We suggest that the knowledge of background region features is more crucial for remote sensing scenes.

Prediction-based distillation utilizes the teacher's predictions as additional objects for the student model and has been widely employed in classification tasks [31], [32], [33]. DML [31] proposes mutual learning to further enhance the student's performance during the distillation process. TAKD [32] introduces teacher assistants to leverage the knowledge from multiple teacher models. DKD [33] focuses on distilling nonobject categories of images, surpassing the performance of feature-based distillation methods. However, prediction-based distillation has been less explored in the context of object detection tasks. Chen et al. [3] initially performed raw distillation on the classification and regression heads in object detection tasks, but it serves as a supplementary approach to feature distillation rather than playing a dominant role. Recently, Zheng et al. [14] reconstructed the process of KD in the regression head to alleviate boundary ambiguity issues. However, the effectiveness of prediction-based distillation still lags behind feature-based distillation. We suggest that this disparity is mainly due to the significant difference in the number of positive and negative samples in detection scenarios, and the gap between lightweight detectors and complex detectors in remote sensing scenes lies in the false alarm rate. Therefore, the knowledge regarding negative samples is more crucial.

III. Method

In this section, we provide a detailed description of our negative-core sample distillation method. Initially, the overall framework of NSD and the training pipeline is presented in Section III-A. Subsequently, we elucidate the innovative components in Sections III-B and III-C, which include the NFA module and the NAP module.

A. Overview Framework

In the detection process, detectors typically preset proposals. The previously mentioned positive and negative samples are classified from these proposals based on the maximum intersection over union (IoU) with all the GTs. Proposals with an IoU greater than a certain threshold are considered positive samples, while those with an IoU below the threshold are considered negative samples.

Previous methods for distilling feature and head knowledge choose samples that are not consistent with the characteristics of large differences in background and target area sizes, as well as similarities in texture and shape, inherent to remote sensing scenarios. This inevitably leads to the omission of some key knowledge or introduced noise. We contend that the key knowledge aligning with remote sensing characteristics is encapsulated within the background response extracted by the teacher model, which significantly differs from those of the student, and in the samples where the student predicts false alarm. Interestingly, negative samples correspond to pixel-level background features at the neck and include logit and a box of student false alarm samples at the head, also encapsulating knowledge of the object to some extent. Thus, we opt for negative samples as our distillation objects. We further define NCSs as the objects of distillation, conducting customized distillation across four levels: pixel, logit, box, and discretized angle.

As illustrated in Fig. 4, we propose an NSD framework comprising an NFA module and an NAP module. The student model, using the NFA module, carefully selects NCS to mimic the negative core features knowledge of the teacher model, which enhances attention to features of regions with similar textures and shapes as the objects. The NAP module, building upon the NCSs selected by the NFA module, conveys category knowledge through a soft supervision mechanism to mitigate the high false alarm rate in the student model, and transfers localization knowledge by decoupling the position and orientation of the distillation bounding box, enabling the student to learn more precise regression and directional predictions. Through the NFA and NAP modules, the carefully selected negative core knowledge of the teacher is transferred to the student model.

The other components of the framework are also detailed in Fig. 4. We use ResNet/Vision Transformer (ViT) as the backbone to extract multiscale features, which are then fused through the FPN. Subsequently, these features are fed into the detection head, which consists of three branches. One branch is the classification head, another is the angle classification head, and the third is the regression head. The angle classification head is specifically designed to predict the angles of bounding boxes independently.

Therefore, the total loss for training the student detector can be represented as follows:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{Ori}} + \lambda_2 \mathcal{L}_{\text{NFA}} + \lambda_3 \mathcal{L}_{\text{NAP}} \tag{1}$$

where $\mathcal{L}_{\text{Ori}}$ represents the original loss of the detector, while $\mathcal{L}_{\text{NFA}}$ and $\mathcal{L}_{\text{NAP}}$, respectively, represent the distillation losses for the features from the neck network and the predictions from the head network. We set $\lambda_1 = \lambda_3 = 1$ and $\lambda_2 = 4$.

B. Negative-Core Sample Feature Assimilation

In this section, the NFA module will be detailed, including the selection of NCS and the student model's imitation of the teacher's negative core features.

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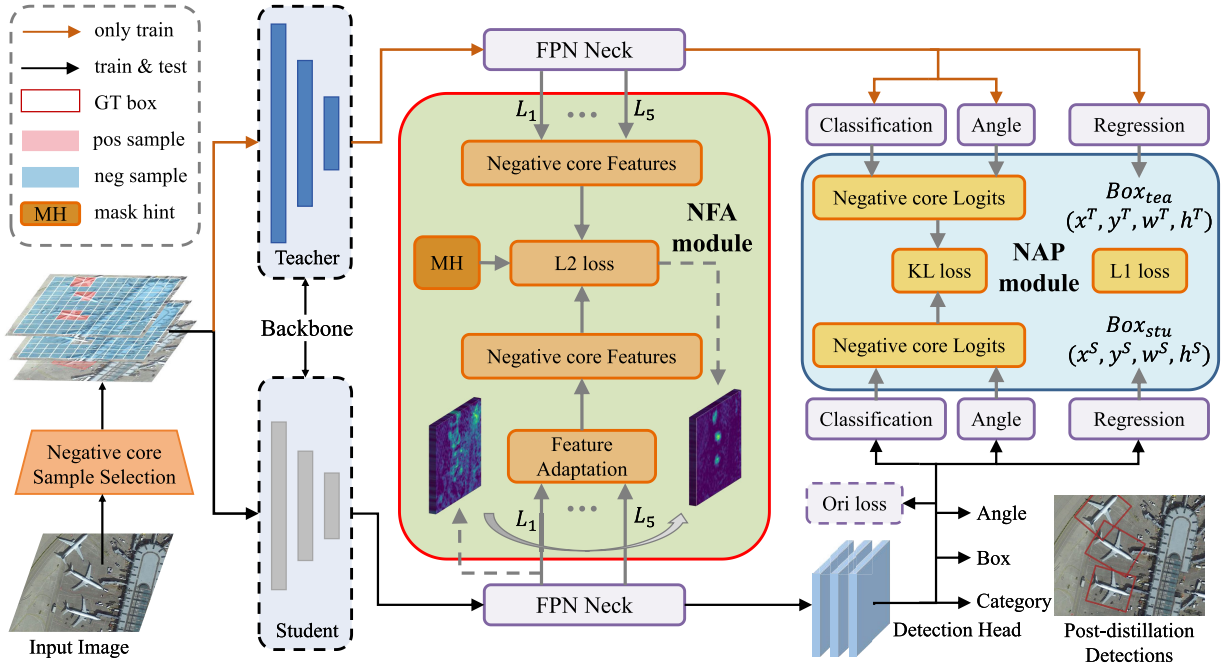


Fig. 4. Framework of our proposed NSD, which comprises the NFA module and the NAP module. The NFA module can be added to the FPN and the NAP module can be adopted for classification and regression in detection head. It can select the negative core knowledge and transfer it from the teacher model to the student model. During the training phase, only the parameters of the student model are updated, and the teacher model is frozen. $(L_1, \ldots, L_5)$ represents the multiple features from the FPN.

1) Negative-Core Samples Selection: In the process of intermediate feature distillation, the student model aims to assimilate key knowledge from the teacher model's features. Previous distillation methods primarily select object regions or positive samples for distillation [5], [34], under the belief that background areas contain noise, leading the model to overlook real objects. However, in remote sensing scenarios, the performance gap between lightweight student models and large teacher models mainly stems from the higher false alarm rate of the student. The distillation of object regions results in students repetitively learning in areas where they already exhibit proficiency, in fact causing them to reiterate knowledge they have previously acquired. This methodology, which mirrors the practice of solely engaging with correct answers while neglecting incorrect ones, yields limited educational outcomes. As a result, the students' high false alarm rates do not improve, and a substantial performance disparity between the students and their instructors remains. Based on these ideas, we define NCS as the objects of distillation to focus more on the area features of the student's false alarms.

The selection of NCS is divided into two parts. The first part focuses on the object aspect, obtained based on the maximum IoU between the preset proposals and the GT of the objects. The second part focuses on the background aspect, which is obtained based on the predictions made by the student model.

Given an input image with a width of W and a height of H, to fully utilize the knowledge of feature maps at various scales, we propose five scales of proposals corresponding to the five scales of features output by the detector's backbone. These are set up with different strides. The proposals are denoted as $\mathcal{A}_k = (a_1^k, a_2^k, a_3^k, a_4^k, a_5^k)$, where $k \in [1, K]$, and K represents the number of types of preset proposals, such as proposals of different scales and aspect ratios.

We map the proposals from the five scales to the corresponding scales of the original image, obtaining $\mathcal{P}_m$, where $m \in [1, M]$ and M represents the total number of proposals. Then, we calculate the IoU between each proposal and all N GT in the image and take the maximum value as the IoU for the proposal. This gives us the IoU value for the mth proposal, denoted as I_m

$$I_m = \max_{1<n<N} (\text{IoU}(\mathcal{P}_m, \mathcal{G}_n)) \tag{2}$$

where $\mathcal{G}_n$ represents the nth GT. With the obtained IoU of the mth proposal, we generate a mask $\mathcal{M}_1^m$

$$\mathcal{M}_1^m = \frac{I_m}{I} \cdot \mathbf{1}_{(0,I)}(I_m), \quad m \in [1, M] \tag{3}$$

where I is set as the IoU threshold, serving as the boundary to distinguish between positive and negative samples. Compared to positive samples where $I_m \in (I, 1)$, negative samples with $I_m \in (0, I)$ pay more attention to the object's edge features. Therefore, negative samples with $I_m \in (0, I)$ will be treated as the first part of the NCS and will be assigned weights proportional to their respective I_m values.

Furthermore, the detector assigns true value labels to the preset M proposals. The GT label for the mth proposal is denoted as y_m

$$y_m = c, \quad c \in [0, 1, \ldots, C_r] \tag{4}$$

where the set $\{0, 1, \ldots, C_r - 1\}$ represents the C_r categories of objects and C_r denotes the background category.

The detector's head predicts the category of the mth proposal as l_m. We select the proposals that are false alarms of the

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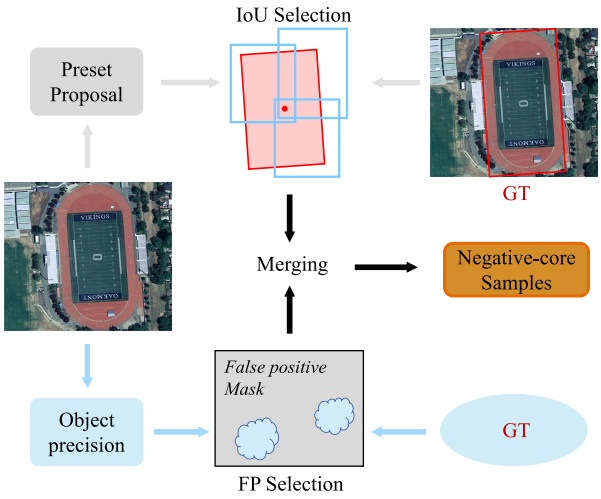


Fig. 5. Process of NCS selection. (Top) First part of NCS. (Bottom) Second part of NCS. "GT" refers to the GT labels assigned to each proposal based on the IoU.

detector as the second part of the NCS. Utilizing the prediction results, we generate a mask $\mathcal{M}_2^m$

$$\mathcal{M}_2^m = s_m \cdot \mathbf{1}_{\{l_m \neq C_r\}}, \quad m \in [1, M] \tag{5}$$

where s_m is the confidence of the mth proposal, which will be used as the weight of the second part of NCS.

So, we finally generated a mask $\mathcal{M}_m$ of NCS

$$\mathcal{M}_m = \mathcal{M}_1^m \bigcup \mathcal{M}_2^m, \quad m \in [1, M]. \tag{6}$$

The process of NCS picking is shown in Fig. 5, and we utilize $\mathcal{M}_m$ to choose NCS.

2) Negative-Core Sample Feature Distillation: To fully assimilate the teacher's intermediate feature knowledge, we approximate the alignment of the teacher–student distribution by minimizing the negative sample feature distillation loss. As the NCS differ across images, we set a normalization coefficient N_m to treat different samples equitably

$$\mathcal{N}_k = \max\left(1, \sum_{m=1}^{M} \lceil \mathcal{M}_m \rceil\right), \quad \mathcal{P}_m \in \mathcal{A}_k \tag{7}$$

where k represents the kth type of proposal shape. We average the distillation losses corresponding to each shape of the proposal. Since we are using the L2 loss, we process the normalization coefficient by taking its half to balance the gradients. The NFA loss can be expressed as

$$\mathcal{L}_{\text{NFA}} = \frac{1}{K} \sum_{k=1}^{K} \frac{1}{2N_k} L_{fk} \tag{8}$$

$$L_{fk} = \frac{1}{CHW} \sum_{c=1}^{C} \sum_{h=1}^{H} \sum_{w=1}^{W} \mathcal{M}_{c,h,w} \left(\varphi\left(F_{c,h,w}^S\right) - F_{c,h,w}^T\right)^2 \tag{9}$$

where φ is the adaptation layer that matches the feature of NCS of the student and teacher models, as shown in Fig. 6. K is the number of shape categories for the preset proposal boxes, used to control the scale of the feature distillation loss. $\mathcal{M}_{c,h,w}$ is the mask $\mathcal{M}_m$ that is mapped onto the image feature.

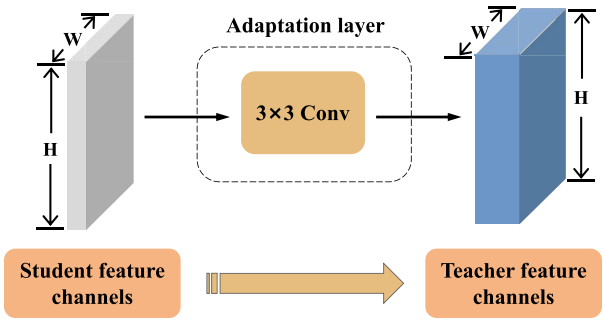


Fig. 6. Adaptation layer. The student feature map channels are equal to the teacher after the adaptation layer.

C. Negative-Core Sample Anti-False Alarm Prediction

In this section, the NAP module will be detailed. We present a distillation approach for the prediction outputs of the head network, which includes distillation of the predicted NCS categories, bounding box positions, and angles.

1) Negative-Core Sample Classification Distillation: Previous methods for distilling the knowledge of object category logit in head outputs primarily borrow from classification tasks and involved transferring knowledge by calculating the KL divergence between the soft logit of students and teachers on positive samples. However, this approach does not yield effective results in detection tasks.

In detection tasks, an image often contains multiple objects. In remote sensing scenarios, the object area is significantly smaller than the background area. This means that when distilling soft logit for positive proposals, the KL divergence loss between the student and the teacher on the object's category logit can miss many key pieces of knowledge from the teacher, especially the knowledge contained in the logit of proposals where the student model has false alarms. As a result, this leads to a limited improvement in the student's classification capabilities.

The student's prediction accuracy on positive proposals is nearly on par with that of the teacher, but significantly lower on negative proposals. The NCS selected in the NFA module addresses this issue. On the one hand, they include the teacher's soft logit for the objects; on the other hand, they contain the teacher's soft logit for the proposals where the student model has false alarms. Therefore, we perform the distillation of soft logit based on NCS.

Regarding the predictions of the detector, for a training sample, the classification probability is denoted as $\mathbf{p} = [p_0, p_1, \ldots, p_{Cr}] \in \mathbf{R}^{\mathbf{1}\times\mathbf{Cr}}$, where p_i is the probability for category i, and Cr is the total number of categories. Each element in $\mathbf{p}$ can be obtained through the softmax function. We do not calculate the background category

$$p_i = \frac{\exp(z_i)}{\sum_{j=0}^{Cr-1} \exp(z_j)} \tag{10}$$

where z_i represents the logit of category i.

Through distilling NCS, the student comprehensively mimics the teacher's core classification knowledge for both objects and backgrounds. We still use the classic KL divergence as the

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distillation loss function

$$L_{\text{cls}} = \sum_{m=1}^{M} \mathcal{M}_m \text{KL}\left(\mathbf{p_m^S} || \mathbf{p_m^T}\right) = \frac{\alpha}{N} \sum_{m=1}^{M} \sum_{i=1}^{Cr} \mathcal{M}_m p_{m,i}^{T} \left(\log\left(p_{m,i}^{T}\right) - \log\left(p_{m,i}^{S}\right)\right) \quad (11)$$

$$N = \max\left(1, \sum_{m=1}^{M} \lceil \mathcal{M}_m \rceil\right) \quad (12)$$

where $\mathbf{p_m^s}$ and $\mathbf{p_m^t}$, respectively, represent the classification probabilities predicted by the student and teacher models for the mth sample, and α is used to control the scale of the category distillation loss.

2) Negative-Core Sample Localization Distillation: To enhance the student's localization capability, localization distillation (LD) methods typically minimize the differences between the bounding box position and shape predictions of the student and teacher models for the objects. This allows the transfer of the teacher's localization knowledge to the student. However, unlike soft predictions, the orientation bounding boxes predicted by the teacher for positive samples do not differ significantly from the GT. They essentially contain the same information and serve as identical supervisory signals for the student. As a result, they do not effectively guide the student in updating parameters toward better localization and therefore fail to efficiently transfer the teacher's localization knowledge.

For the NCS selected by the NFA module, the teacher's regression results may not be identical to the GT, which can provide additional localization knowledge to the student to assist in parameter updates. We refer to this as "dark knowledge" for localization.

Rotated bounding boxes are typically represented using a five-parameter notation. Let $\mathbf{b_m^s}$ and $\mathbf{b_m^t}$, respectively, represent the rotated bounding boxes predicted by the student and the teacher for the mth proposal

$$\mathbf{b_m^s} = \left(x_m^s, y_m^s, w_m^s, h_m^s, \theta_m^s\right)$$
$$\mathbf{b_m^t} = \left(x_m^t, y_m^t, w_m^t, h_m^t, \theta_m^t\right) \quad (13)$$

where x and y are the horizontal and vertical coordinates of the center point of the rotated bounding box, respectively, w and h represent the length and width, respectively, and θ is the angle difference between the rotated bounding box and the preset proposal.

Due to the complex periodic nature of angles [15], [35], the LD component focuses exclusively on the distillation of the first four coordinates. The distillation loss for the first four regression parameters of the bounding box is defined as follows:

$$L_{\text{loc}} = \frac{\beta}{N} \sum_{m=1}^{M} \mathcal{M}_m \left(\mathbf{b_m^s} - \mathbf{b_m^t}\right) \quad (14)$$

where β is used to control the scale of the LD loss.

3) Negative-Core Sample Angle Distillation: For orientation bounding boxes, the definition of the angle is dependent on the reference of the initial direction or the representation of the box edges. For the typical definition of a long edge with le90, due to the periodic nature of the angle, an orientation bounding box with an angle ψ is equivalent to one with $\psi + \mu \times 180°$, where $\mu = 1, \ldots, n$. To avoid ambiguity in the direction definition, the range of the angle is usually set as $[-90°, 90°)$. Since the model predicts the angle continuously, during the distillation process, the training loss is increased by the boundary discontinuity when the predicted angle crosses the range boundary in adjacent training steps.

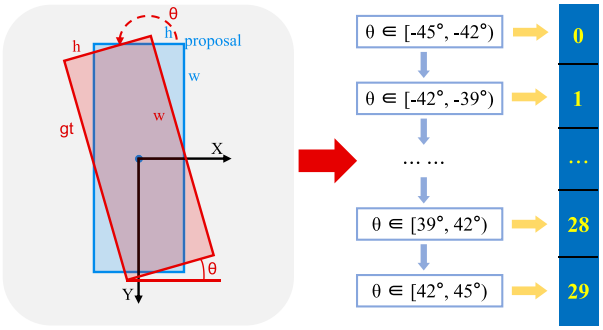


Fig. 7. Flow of angle coding for a given angle difference with 30 categories.

During the process of distilling angles, discontinuities at the boundaries can lead to a stepwise increase in sample loss. With the quantity of NCS far exceeding that of positive samples, the loss further exacerbates in a geometric fashion, which leads to errors in angle learning. To circumvent this issue, we reframe the continuous angle prediction regression task as a classification task, thereby distilling angles through categorization.

Let the angle for the preset proposal be θ_p, and the corresponding angle of GT be θ_g. We transform the angle difference between them in such a way that the final regression angle difference θ is always minimized. First, we constrain θ within the original angle range of θ_g. Since the bounding box has different angles, when w and h with different edges are two values of θ

$$d\theta_1 = (\theta_p - \theta_g + 90°) \bmod 90 - 45°$$
$$d\theta_2 = (\theta_p - \theta_g + 180°) \bmod 90 - 45°. \quad (15)$$

To ensure that θ is always minimum, therefore θ is defined as

$$\theta = \min(d\theta_1, d\theta_2). \quad (16)$$

After the transformation, $\theta \in [-45°, 45°)$. We divide it into AL_{cls} categories, as shown in Fig. 7, we set $\text{AL}_{\text{cls}} = 30$, each containing a range of 3°. We employ angle encoding and decoding at distinct phases of training and testing. During training, angles are encoded into discrete categories to circumvent abrupt changes in loss, whereas, at the testing phase, the categories of angles are decoded back into their true angles to compute the IoU with the GT. Suppose the angle belongs to the ζth category, the encoding and decoding process of the angle is as follows:

$$\zeta_m = \begin{cases} (\theta_m + \theta_{\text{offset}})/3, & \text{if } \mathcal{M}_m > 0 \\ 30, & \text{otherwise} \end{cases} \quad (17)$$

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TABLE I
ABLATION STUDY AND EVALUATIONS OF THE PROPOSED NSD METHOD UNDER MAP (%) ON THE DOTA DATASET

Modules	NFA	NAP			BD	BR	GTF	LV	BC	SBF	HA	SP	HC	mAP	mAP_{75}
		CD	AD	LD											
Teacher					79.57	41.53	66.86	63.84	82.34	57.04	59.32	66.26	46.77	69.68	41.88
Selected Modules					72.80	35.37	57.64	55.67	76.92	45.00	49.83	56.80	32.70	63.52	37.18
			✓		75.40	36.91	63.62	57.50	76.34	50.20	50.89	64.38	38.37	66.00(+2.48)	38.37(+1.19)
				✓	76.27	39.22	69.77	61.88	80.45	54.36	56.16	65.52	38.55	67.86(+4.34)	40.11(+2.93)
		✓			80.89	38.92	69.46	61.54	81.35	54.25	52.99	67.91	43.19	68.23(+4.71)	38.15(+0.97)
	✓				77.57	38.39	67.56	64.97	81.13	53.85	55.27	65.85	44.05	68.36(+4.84)	40.34(+3.16)
		✓	✓		79.93	38.69	70.27	62.38	82.52	57.34	54.77	67.90	45.59	69.03(+5.51)	39.28(+2.10)
		✓	✓	✓	82.86	39.63	71.19	64.43	81.00	57.89	55.30	67.50	44.39	69.45(+5.93)	40.52(+3.34)
	✓	✓	✓	✓	82.05	38.94	71.29	65.10	83.06	56.41	57.23	67.52	44.81	69.87(**+6.35**)	40.78(**+3.60**)

where θ_m represents the angle difference for the mth sample, and θ_{offset} is the angle offset to ensure that the angle is a nonnegative value. During the decoding process of the angle, we use the midpoint of the predicted category interval as the final predicted angle

$$\theta_m = (\zeta_m + 0.5) \times 3 - \theta_{\text{offset}}. \tag{18}$$

The decoding process of the angle may introduce some precision bias. When a category represents an interval range of 3°, the maximum precision loss for the angle is 1.5°, with an expected precision loss of 0.75°. Even for rectangles with an aspect ratio of 1:9, the negative effect can be considered negligible [15].

For the distillation of predicted angle categories, we follow the same distillation process as the NCS classification distillation (CD). They adhere to a unified distillation paradigm. Consequently, we utilize a similar probability distribution, denoted as $\mathbf{q} = [q_0, q_1, \ldots, q_{30}] \in \mathbf{R}^{\mathbf{1}\times\mathbf{30}}$, to represent the output probabilities of angle categories and employ the same loss function to calculate the loss.

Our distillation process is as follows:

$$\begin{aligned} L_{\text{angle}} &= \sum_{m=1}^{M} \mathcal{M}_i \text{KL}\left(\mathbf{q}_\mathbf{m}^\mathbf{S}, \mathbf{q}_\mathbf{m}^\mathbf{T}\right) \\ &= \frac{\gamma}{N} \sum_{m=1}^{M} \sum_{j=1}^{30} \mathcal{M}_m q_{i,j}^T \left(\log\left(q_{m,j}^T\right) - \log\left(q_{m,j}^S\right)\right) \end{aligned} \tag{19}$$

where $\mathbf{q}_\mathbf{m}^\mathbf{s}$ and $\mathbf{q}_\mathbf{m}^\mathbf{t}$, respectively, represent the classification probabilities of the discretized angle predicted by the student and teacher models for the mth sample. γ is used to control the scale of the angle distillation (AD) loss.

In summary, the total distillation loss for the predicted outputs of the detector's head network is as follows:

$$\mathcal{L}_{\text{NAP}} = L_{\text{cls}} + L_{\text{loc}} + L_{\text{angle}}. \tag{20}$$

IV. EXPERIMENTS

A. Experimental Setup

1) Datasets: To evaluate the effectiveness of our proposed NSD framework, we have conducted comprehensive experiments using well-known publicly available aerial imagery datasets, including DOTA [16], DIOR [17], and HRSC2016 [18], all of which provide annotations in OBBs.

DOTA: The DOTA dataset [16], an established benchmark for object detection in aerial imagery, comprises three different versions: DOTAv1.0, DOTAv1.5, and DOTAv2.0. In this work, we utilize DOTAv1.0 in our experiments, which encompasses a compilation of 2806 images sourced from various remote sensing platforms. This dataset is characterized by a rich diversity, with 188 282 instances spanning 15 distinct scene categories. Image dimensions within this dataset vary significantly, ranging from 800 × 800 to a substantial 4000 × 4000 pixels. For our experiments, we segment these images into 1024 × 1024 pixels, incorporating a 200-pixel overlap to ensure comprehensive coverage. The partitioning of the dataset into trainval and test subsets is executed at a ratio of 2:1.

DIOR: The DIOR dataset [17] boasts a total of 23 463 images, and 190 288 annotated objects for optical remote sensing object detection, covering an array of 20 scene categories. Each image maintains a resolution of 800 × 800 pixels. For experimental consistency, we divide these images into subimages with a 200-pixel overlap, mirroring the original image size. The dataset is bifurcated into a trainval set of 11 725 images and a test set comprising 11 738 images.

HRSC2016: The HRSC2016 dataset [18] consists of 1061 high-resolution images annotated with OBB for the detection of ships, totaling 1228 valid annotated instances. The dimensions of the images are notably varied, ranging from 300 × 300 to 1500 × 900 pixels. In our experimental setup, we standardize the image size to 800 × 512 pixels to maintain uniformity. The dataset is divided into a trainval set with 617 images and a test set containing 453 images.

2) Evaluation Metric: In Aerial Object Detection, mean AP (mAP) at an IoU threshold of 0.5 is the standard metric for evaluating detection algorithms. In our experiments, we use mAP to assess overall detection performance across classes. To further highlight the localization improvements of our student model, we also considered mAP_{75}, which measures precision at a stricter IoU threshold of 0.75, and $mAP_{50:95}$, which averages precision across IoU thresholds from 0.5 to 0.95, offering a comprehensive evaluation of the model's performance at varying detection difficulty levels.

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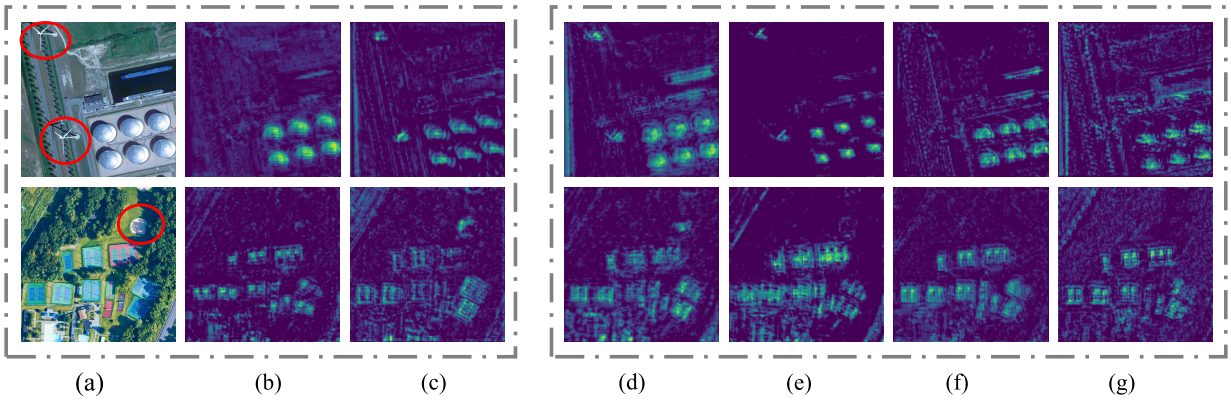


Fig. 8. Compared feature visualization results between the baseline and our NFA model with different weights. Our NFA model enhances feature extraction in object regions and significantly improves the accuracy of background feature delineation in nonobject areas, surpassing the baseline performance. On the right side of the image, a detailed comparison is provided to illustrate the impact of our proposed NCS feature methodology on the student features under varying weight conditions, where BaseDistill denotes the distillation of object feature region. (a) Ori image. (b) Teacher. (c) Student. (d) BaseDistill. (e) NFA ($\lambda = 4$). (f) NFA ($\lambda = 12$). (g) NFA ($\lambda = 20$).

TABLE II
ABLATION STUDY OF THE PROPOSED NFA MODULE FOR WEIGHT (λ) ON THE DOTA DATASET

Methods	λ	mAP	mAP_{75}	$mAP_{50:95}$
BaseDistill	-	66.72	37.91	37.83
Negative-core sample Feature Assimilation	4	65.39	37.40	37.12
	8	67.38	38.70	38.42
	12	68.36	40.34	39.37
	16	68.54	39.99	39.40
	20	67.32	39.13	38.61

TABLE III
ABLATION STUDY OF THE PROPOSED NAP MODULE FOR TEMPERATURE (τ) ON THE DOTA DATASET

Methods	τ	mAP	mAP_{75}	$mAP_{50:95}$	mAoE↓
Baseline	-	63.52	37.18	36.48	6.97
Cls + Angle Distillation	1	68.40	38.73	39.09	7.00
	2	68.64	39.91	39.22	6.81
	4	69.03	39.28	39.28	6.57
	8	68.44	39.93	39.25	6.85
	16	68.30	39.89	39.22	6.80

In addition, we have integrated the mean average orientation error (mAoE°), which serves to quantify the angular discrepancy in the predicted orientation, thereby providing insights into the directional accuracy of the model.

In the assessment of the model's efficiency, we utilize frames per second (FPS), model parameters (Params), and floating-point operations (FLOPs). Parameters reflect the model's magnitude, denoting the cumulative tally of parameters across the model's architecture, including the backbone, neck, and detection head. FLOPs offer a gauge of the model's computational complexity, denoting the number of such operations required by the model. The metric of FPS is deployed during the inference phase to ascertain the model's expedience, indicative of the volume of imagery the model is capable of processing each second.

3) Implementation Details: Our experimental protocol is established within the mmRotate framework [36], which includes both single-stage and two-stage detectors reliant on anchor boxes, thus covering a comprehensive range of architectural paradigms. The backbone networks of these models are either ResNet or Transformer. The distillation technique is only confined to the training phase, where model parameters are refined using the stochastic gradient descent (SGD) optimizer. This process adheres to the learning rate specifications intrinsic to each model's original framework, thereby preserving the intended dynamics of the learning algorithm. We have standardized the momentum and weight decay parameters across all models to 0.9 and 0.0001, respectively, to maintain consistency in the optimization process.

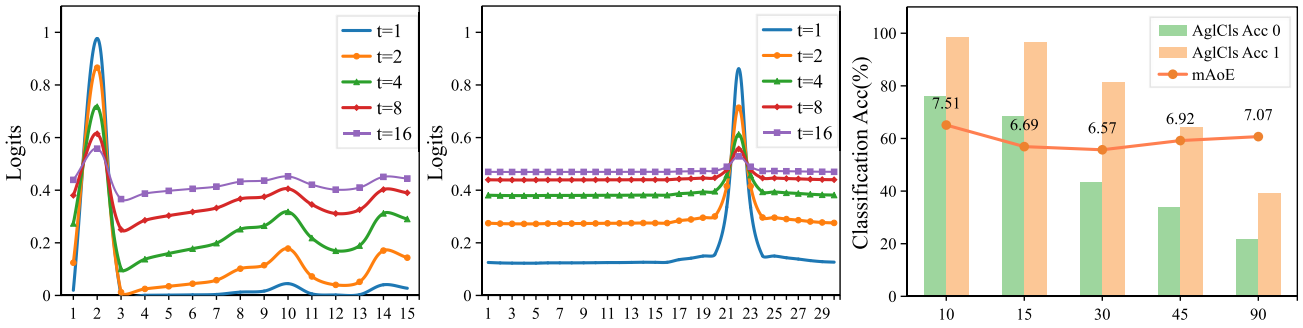
Training sessions are conducted on an RTX 3090 24-GB GPU, accommodating a batch size of two to optimize the resource-to-performance ratio. The training regimen for the HRSC2016 dataset spans 72 epochs to ensure thorough learning. Conversely, for the more voluminous datasets, DOTA and DIOR, the training duration is condensed to 12 epochs.

B. Ablation Experiment

In the ablation study, a series of experiments are designed to ascertain the efficacy of the proposed method. We incorporate ResNet18 and ResNet101 as the backbone for the student and teacher models. Moreover, RetinaNet(OBB) is selected as the detection framework due to its widespread adoption in object detection tasks. To facilitate a comprehensive and rigorous assessment, we employ the DOTA datasets, which are known for their extensive coverage and heterogeneity. Table I delineates the outcomes derived from a series of eight experiments, elucidating the contribution of individual and combined components to the overarching performance of the model.

1) Evaluation of NFA: In this study, we visualize the feature maps of the student model in Fig. 8. These images reveal that the original student model differs from the original image to varying degrees in both the object and background regions. Notably, the differences in the background regions are even

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(a) Cls Category (b) Angle Category (c) Angle Lable Number

Fig. 9. (a) and (b) Visualization of the distribution of classification soft logit and angle category soft logit of the teacher model under different temperature conditions τ. (c) Angle prediction of the model across varying numbers of categories. The green bars represent the angle classification accuracy when the categorization is entirely correct, while the yellow bars indicate the angle classification accuracy with an error margin not exceeding adjacent categories. The line graph represents the mAoE metric.

TABLE IV
COMPARISON RESULTS OF DIFFERENT NUMBERS OF ANGLE LABEL (AL_{cls}) ON THE DOTA DATASET

Methods	AL_{cls}	mAP	mAP_{75}	$mAP_{50:95}$	mAoE↓
Baseline	-	63.52	37.18	36.48	6.97
Cls + Angle Distillation	10	68.89	39.85	38.97	7.51
	15	68.63	39.74	39.28	6.69
	30	69.03	39.28	39.28	6.57
	45	68.06	39.75	39.07	6.92
	90	68.48	38.74	38.74	7.07

TABLE V
ANALYSIS OF DIFFERENT HYPERPARAMETERS IN THE NAP MODULE ON THE DOTA DATASET

α	β mAP			
	0.5	1.0	1.5	2.0
1.0	68.9	68.9	69.2	69.0
2.0	69.1	69.5	69.3	69.0
3.0	69.0	69.2	69.0	68.9

TABLE VI
ANALYSIS OF DIFFERENT HYPERPARAMETERS BETWEEN THE NFA MODULE AND THE NAP MODULE ON THE DOTA DATASET

λ_2	λ_3 mAP			
	0.5	1.0	1.5	2.0
1.0	68.2	69.6	69.7	69.7
4.0	68.8	69.9	69.3	69.4
8.0	69.0	69.9	69.8	69.6

more pronounced than in the object regions, directly leading to the student model's misclassification of some background elements as objects. After basic feature distillation [5], the student model aligns more closely with the original image in terms of object region features, yet the issue of false alarms is not mitigated. The integration of the NFA module substantially improves this phenomenon.

TABLE VII
COMPARISON RESULTS OF DISTILLATION OF DIFFERENT NEGATIVE SAMPLES

Modules	-	Neg_1	Neg_2	Neg_3	Neg_4(NCS)
mAP	63.5	68.6	68.2	68.9	69.9

Following the distillation of pixel-level features corresponding to NCS, the student model's performance metrics improve significantly. Serving as a structure at the neck level, the NFA module conveys semantic and positional information contained in the NCS from the teacher model to the student. As shown in Table I, with the aid of the NFA module, the mAP and mAP_{75} values of the student model increase by 4.84% and 3.16%, respectively. Remarkable performance improvements are also observed in categories prone to false alarms by the student model, such as "ground track field" (GTF) and "swimming pool," with their AP values increasing by 9.92% and 9.05%, respectively.

The enhancement of the student model's features by the NFA module under varying loss weights is demonstrated in Fig. 8. As the weight increases, the student model's features converge comprehensively toward those of the teacher model. Within a certain weight range, the valuable knowledge of objects and background carried by the NCS features aids in performance improvement for the student model, as presented in Table II. However, outside this range, performance begins to decline relative to lower weights due to the introduction of excessive noise from overly large weights, resulting in a deterioration of the student model's performance, which, nonetheless, is still superior to that achieved through basic feature distillation.

2) Evaluation of NAP: To investigate the impact of various components within the NAP module on model performance, we conduct separate experiments for the CD module, AD nodule, and LD module.

Impact of CD and AD: As integral parts of the NAP module, CD and AD share a similar structure, and both utilize the knowledge of soft logit from the teacher model, so we conduct a unified ablation experiment on AD and CD. Due to the inevitable noise within the knowledge carried by the

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TABLE VIII

COMPARISON RESULTS OF DISTILLATION OF POSITIVE SAMPLES AND PROPOSED NCS. WE LIST THE MAP (%) UNDER DIFFERENT IOU THREADS IN EACH MODULE. WHERE "-" REPRESENTS THE ORIGINAL STUDENT METHOD WITHOUT ANY DISTILLATION

Modules	NFA			NAP:CD			NAP:LD			NAP			NSD		
	-	Pos	NCS	-	Pos	NCS	-	Pos	NCS	-	Pos	NCS	-	Pos	NCS
mAP	63.52	66.72	68.36	63.52	65.56	68.23	63.52	63.90	67.86	63.52	65.84	69.45	63.52	66.00	69.87
mAP_{75}	37.18	37.91	40.34	37.18	36.92	38.15	37.18	36.83	40.11	37.18	37.26	40.52	37.18	37.57	40.78
$mAP_{50:95}$	36.48	37.83	39.37	36.48	36.53	38.27	36.48	36.32	39.03	36.48	36.88	39.83	36.48	37.17	39.88

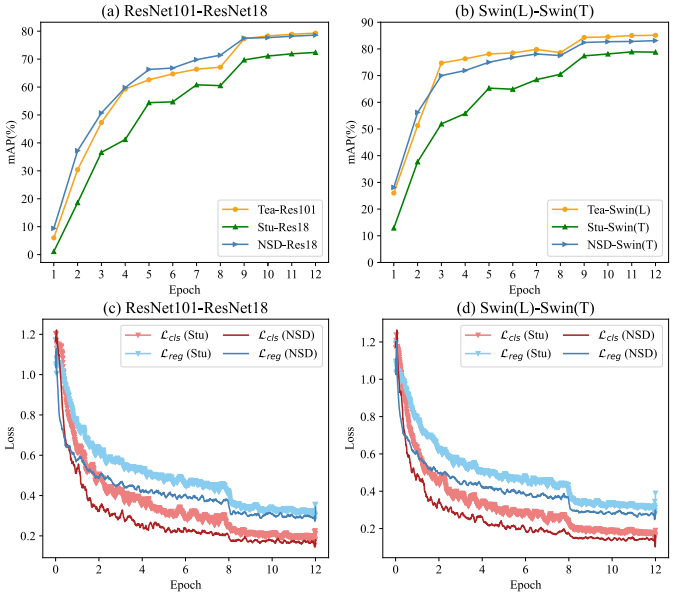


Fig. 10. (a) and (b) Comparison of mAP during the training process among the teacher, the predistillation student, and the postdistillation student facilitated by the NSD method, based on ResNet and Swin Transformer backbones, respectively. (c) and (d) Comparison of loss during the training process between the predistillation student and the postdistillation student based on ResNet and Swin Transformer backbones, respectively.

TABLE IX

EFFICIENCY OF RETINANET SERIES NETWORKS BEFORE AND AFTER OD ON THE DOTA DATASET. 18 AND 101 REPRESENT THE NUMBER OF LAYERS OF THE BACKBONE IN THE DETECTOR

Method	FLOPs(G)	Params(M)	FPS(img/s)	mAP
RetinaNet Res101	293.81	55.41	19.5	69.68
RetinaNet Res18	177.75	20.55	29.6	63.52
NSD Res18	177.75	20.55	29.4	69.87
RetinaNet Swin-L†	893.50	207.34	6.5	76.62
RetinaNet Swin-T	235.68	37.75	18.5	69.80
NSD Swin-T	235.68	37.75	18.7	73.90

Note: The results presented are all based on tests conducted within the mmrotate framework on a single RTX 3090 24GB GPU.

CD and AD modules, we incorporate a warmup strategy during model training, omitting additional supervisory signals in the first training iteration and gradually increasing the teacher's supervisory signals in the second and third iterations. As shown in Table I, with the introduction of CD, the mAP and mAP_{75} metrics of the student model improve by 4.77% and 0.97%, respectively, almost matching the assistance provided by the NFA module and confirming the significance of logit as advanced semantic knowledge. Further inclusion of AD

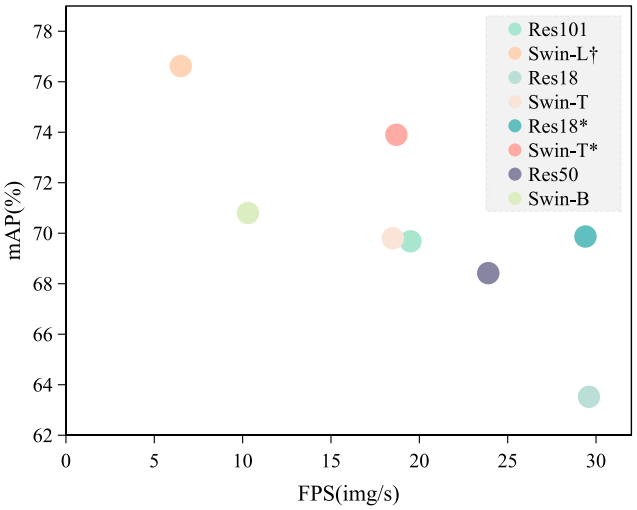


Fig. 11. Comparisons of the teacher and lightweight student model in terms of FPS (img/s) and mAP, based on ResNet and Swin Transformer, respectively. "∗" represents the model after distillation with our NSD method.

leads to additional increases of 0.34% and 0.95% on the mAP and mAP_{75} metrics, respectively, based on CD, achieving further performance improvements in categories with large aspect ratios and near-square shapes, such as "harbor" and "helicopter" (HL), which see AP values increase by 1.71% and 3.43%, respectively.

We also analyze the performance enhancement of the student model under varying temperatures of the teacher model's logit knowledge, as shown in Table III. An appropriate temperature can soften the teacher's logit, conveying more valuable information. As the temperature rises, the student model's performance further improves. However, excessively high temperatures render the soft labels provided by the teacher model too diffuse, as shown in Fig. 9(a) and (b), failing to provide accurate category and orientation information, which can lead to a decline in model performance instead of improvement.

We further analyze the performance differences under different numbers of angle classifications, especially the mAoE metric. As shown in Fig. 9(c), the number of angle categories directly affects the accuracy of orientation prediction. Too few angle categories make the model's classification results more accurate but with a larger error due to the broader angle range per category. As the number of angle classes increases, the objective error caused by each category's angle range decreases, but this increases the model's classification difficulty, leading to a drop in the classification accuracy. When the difficulty of angle classification is balanced with the

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TABLE X
RESULTS OF MORE DETECTORS WITH DIFFERENT BACKBONES ON THE DOTA DATASET. † MEANS USING MULTISCALE TRAINING. SWIN-B AND SWIN-T REPRESENT THAT THE SCALE OF SWIN TRANSFORMER ARE BASE AND LARGE, RESPECTIVELY

Student	Teacher(B)	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP	mAP_{75}
R-RetinaNet ResNet18 [20]	-	88.9	72.8	35.4	57.6	73.6	55.7	76.6	90.9	76.9	80.2	45.0	60.0	49.8	56.8	32.7	63.5	37.2
	ResNet101	89.3	82.1	38.9	71.3	77.0	65.1	78.0	90.9	83.1	81.5	56.4	64.9	57.2	67.5	44.8	69.9	40.8
	ResNeXt101	89.2	80.3	39.9	73.0	76.5	64.6	77.6	90.9	81.5	82.5	57.6	63.8	56.4	68.1	39.8	69.5	40.7
	Swin-B	89.5	78.9	39.0	72.4	76.9	64.4	77.8	90.8	82.4	83.1	55.1	63.1	56.3	69.7	52.5	70.1	40.3
	Swin-L	89.5	80.4	39.6	71.8	77.2	65.5	77.8	90.9	84.5	83.4	60.7	63.2	56.9	70.6	43.9	70.4	40.2
R-Faster RCNN ResNet18 [22]	-	89.2	77.9	46.4	64.9	76.4	73.5	85.1	90.9	83.2	84.6	50.5	59.9	64.4	66.8	58.4	71.5	39.4
	ResNet101	89.7	80.9	46.1	67.8	76.4	73.1	85.7	90.9	85.1	83.9	58.0	62.8	65.6	66.0	55.1	72.5	41.0
	ResNeXt101	89.3	81.9	48.2	67.8	76.1	74.2	85.9	90.9	83.6	84.4	54.9	64.0	66.1	67.2	51.8	72.4	42.2
	Swin-B	89.4	80.9	49.0	70.2	76.6	74.3	85.3	90.9	85.2	85.2	57.3	63.1	65.6	66.7	54.9	73.0	40.3
	Swin-L	89.6	83.2	48.5	68.2	77.1	74.9	85.4	90.9	84.9	84.3	58.1	63.1	65.7	68.8	53.2	73.1	40.6
R-ATSS ResNet18 [37]	-	88.7	71.0	43.3	65.1	75.2	68.4	76.2	90.9	82.5	83.4	49.4	58.9	59.4	64.3	35.5	67.5	37.8
	ResNet101	87.7	76.8	47.8	70.8	75.7	73.4	82.5	90.9	79.8	83.1	56.4	64.8	63.7	66.0	48.9	71.2	40.3
	ResNeXt101	88.5	79.6	47.1	70.5	76.7	73.0	82.1	90.9	81.2	83.3	58.6	63.5	63.7	65.6	48.4	71.5	41.6
	Swin-B	89.0	79.0	46.3	71.5	76.2	72.0	82.4	90.9	82.7	84.5	56.4	63.6	63.5	66.3	56.7	72.1	40.1
	Swin-L	88.9	80.2	47.0	70.9	76.2	72.1	82.4	90.8	86.2	84.9	61.8	63.1	64.0	67.6	60.4	73.1	40.8
R-RetinaNet Swin-T [38]	-	89.4	79.7	41.1	71.7	77.2	64.5	77.9	90.9	84.9	84.0	56.1	63.5	52.8	70.4	43.0	69.8	39.4
	ResNet101†	89.6	83.0	43.4	76.5	77.5	69.1	82.4	90.9	83.0	83.2	63.5	66.1	62.2	71.0	56.3	73.2	44.4
	ResNeXt101†	90.0	84.6	44.2	77.0	77.7	69.3	82.5	90.8	84.4	84.1	63.7	66.5	62.6	71.2	60.1	73.9	43.7
	Swin-B†	89.6	82.1	42.9	75.5	78.3	69.1	84.0	90.9	86.4	85.1	61.3	69.0	61.7	70.5	59.1	73.7	43.0
	Swin-L†	89.6	84.7	41.3	76.8	78.4	70.0	84.3	90.9	86.4	84.5	60.4	66.8	62.6	71.1	60.8	73.9	44.3

error caused by the angle range of each category, the highest prediction accuracy for orientation is achieved, as shown in Table IV.

Impact of LD: LD uses the knowledge contained in the logit of NCS as an additional supervisory signal to help improve the student model's localization performance. As shown in Table I, after incorporating LD, the student model's mAP and mAP_{75} metrics increase by 4.34% and 2.93%, respectively, particularly achieving surprising performance improvements in categories with significant aspect ratios such as "bridge," "large vehicle" (LV), and "LV."

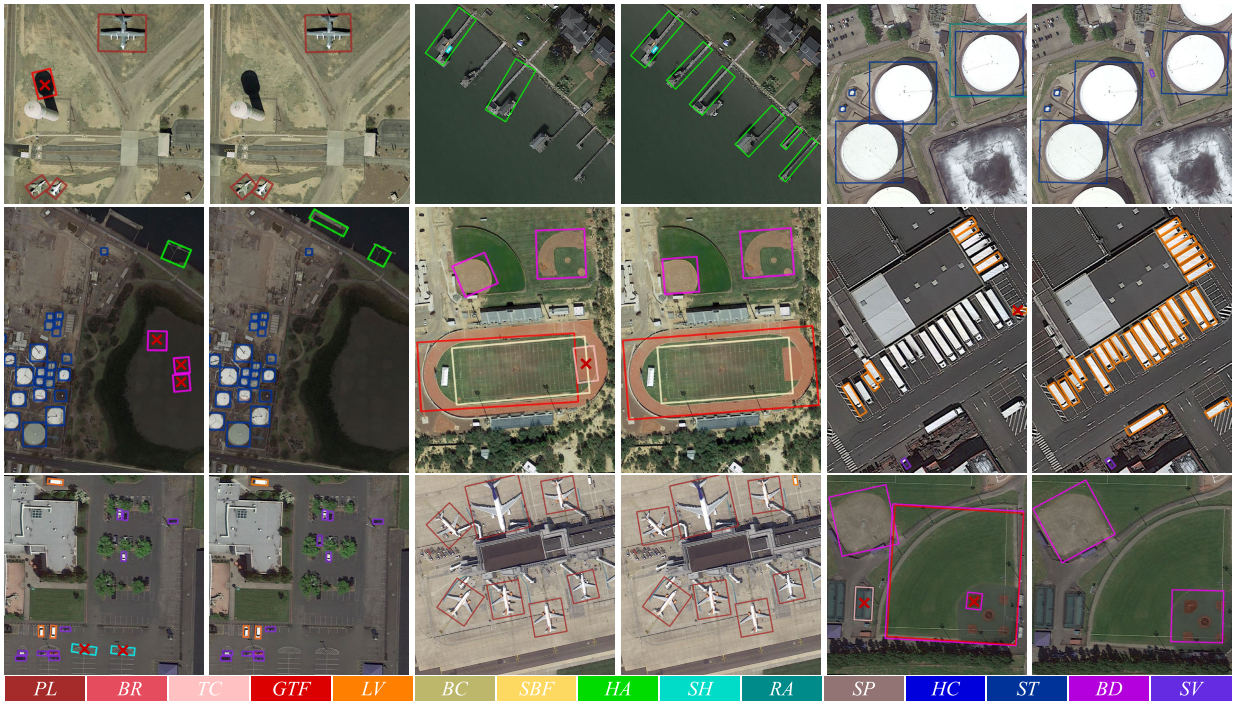
3) Influence of Different Hyperparameters: To validate the stability and optimality of the model, we first analyze the influence of hyperparameters on the NAP module. Since the category of the object and the location of the bounding box directly determine the model's detection performance, we set an appropriate loss weight for AD at 1 and explored the optimal loss weights for CD and LD. Correct prediction of the object category is particularly important, so we set a larger weight adjustment range for CD and began to adjust the weights of both modules. The experimental results of hyperparameters, as shown in Table V, indicate that NAP exhibits robustness within a reasonable range of varying parameters and possesses an optimal weight setting.

In addition, we analyze the impact of the loss weights for the NFA and NAP modules. As shown in Table VI, we conducted experiments with different combinations of loss weights. As the loss weights increased, the model's accuracy gradually improved. However, if the weights of both modules were excessively increased, the model's accuracy experienced some decline. This is because the overly large distillation loss weights reduced the student model's focus on the GT knowledge in the dataset, leading to a loss of accuracy.

4) Comparison of Different Negative Sample Distillation: We analyze the effect of negative sample knowledge on students under different selection conditions and conduct distillation experiments based on four types of negative samples (including our NCS), denoting as Neg_1 [6], Neg_2 [49], Neg_3, and Neg_4 (NCS). Neg_1 refers to the distillation of features in nonobject areas by instance division, Neg_2 refers to the distillation of proposals whose distance from the center of gt is more than the distance from the center to the corner of that gt, Neg_3 refers to the average distillation of proposals whose IoU with gt is in the range $[0, I]$, and Neg_4 refers to the NCSs proposed in this article. As shown in Table VII, all four types of negative samples are effective, proving the role of negative samples. The distillation effect of (NCS) is the best. Compared to (NCS), Neg_1 inevitably introduces more background noise and cannot effectively utilize the high-level semantic and positioning knowledge outputted by the teacher's detection head; Neg_2 has poor adaptability to object size and location due to the center point distance criterion; Neg_3 fails to effectively use the core knowledge in negative samples while introducing noise.

5) Comparison of Positive and Negative-Core Sample Distillation: To demonstrate that the NCS proposed is superior to the positive samples in traditional distillation, we compare the performance of students with knowledge from distilled positive samples and our proposed NCS. Specifically, experiments are established in both the NFA module and the NAP module, corresponding to distillation at the neck feature and the detection head's prediction output, respectively, and

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(a) Baseline (b) NSD (ours) (c) Baseline (d) NSD(ours) (e) Baseline (f) NSD(ours)

Fig. 12. Qualitative comparison results of NSD and the baseline network on the DOTA dataset. The bounding boxes indicate the object detection results. (a), (c), and (e) Visualization results were obtained by the baseline without any distillation. (b), (d), and (f) Results obtained by the baseline distilled through our NSD framework which transfers knowledge of the NCS and replaces the direct angle representation with angel categories. The alarm objects detected by the baseline disappear in NSD's detection. The orientation angles of the object bounding boxes predicted by NSD are more accurate than those procured by the baseline, especially for the long and slim objects. The OBBs detected by NSD shown in (b), (d), and (f) well align the objects.

TABLE XI

COMPARISON RESULTS WITH STATE-OF-THE-ART METHODS ON THE DOTA DATASET. THE ITEMS WITH RED AND BLUE COLORS INDICATE THE BEST AND THE SECOND-BEST RESULTS OF EACH COLUMN, RESPECTIVELY. "†" MEANS USING MULTISCALE TRAINING

Methods	Stage	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP	Speed (FPS)
R-Faster RCNN [22]	Two	89.26	82.40	50.02	69.37	78.17	73.54	85.93	90.90	84.08	85.49	57.58	60.98	66.25	69.23	57.74	73.40	21.4
RoI-Trans [25]	Two	88.97	82.13	54.58	76.28	79.28	77.95	87.94	90.91	87.19	85.64	62.21	62.63	74.63	72.42	59.23	76.13	18.4
Gliding Vertex [39]	Two	89.21	75.77	51.35	69.52	78.10	75.60	86.85	90.90	85.40	84.77	53.50	66.65	66.31	70.00	54.40	73.22	20.6
ReDet [28]	Two	89.20	83.78	52.16	73.31	78.05	82.48	88.23	90.86	87.26	85.95	65.65	62.87	75.89	70.02	66.79	76.83	11.6
Oriented R-CNN [40]	Two	89.36	81.53	52.62	75.03	79.03	82.40	87.82	90.90	86.46	85.30	63.36	65.67	68.28	70.47	57.21	75.69	18.9
ARC-Oriented R-CNN [41]	Two	89.62	81.54	52.59	75.10	79.06	82.29	87.93	90.90	87.11	85.26	63.37	68.73	70.14	70.53	59.36	76.24	16.3
R-RepPoints [42]	Refine	83.36	65.37	36.32	51.62	71.26	51.81	72.45	90.51	70.46	81.97	48.00	58.94	50.68	55.36	3.55	59.44	18.4
R3Det [27]	Refine	89.30	75.21	45.39	69.24	75.54	72.90	79.29	90.89	81.03	83.27	58.82	63.16	63.39	62.21	37.40	69.80	16.7
S2ANet [26]	Refine	89.25	81.07	51.56	71.44	78.60	77.42	86.76	90.90	86.34	84.64	61.30	65.63	66.12	67.69	50.19	73.93	18.3
CFA [43]	Refine	88.04	74.65	49.18	74.59	79.05	74.14	86.73	90.87	80.33	85.95	48.71	58.57	64.40	66.82	22.37	69.63	19.6
SASM [44]	Refine	80.79	66.38	45.41	65.59	72.07	72.35	79.78	90.90	75.69	83.06	48.83	59.19	63.75	59.42	33.19	66.43	18.5
Oriented RepPoints [45]	Refine	87.79	77.66	49.55	66.46	78.37	73.12	86.59	90.87	83.84	84.35	53.19	65.64	63.70	68.72	45.91	71.72	18.1
R-RetinaNet [20]	One	89.30	79.57	41.53	66.86	77.36	63.84	77.92	90.89	82.34	82.86	57.04	63.37	59.32	66.26	46.77	69.68	19.5
R-FCOS [46]	One	89.18	71.93	47.95	61.64	79.30	73.52	85.77	90.90	81.06	84.30	59.57	62.69	62.07	69.94	49.26	71.27	21.4
R-ATSS [37]	One	88.71	76.01	48.67	69.37	76.83	71.35	82.59	90.87	77.68	83.80	57.92	66.24	64.26	65.34	39.85	70.63	22.2
CSL [15]	One	89.34	79.71	40.84	69.93	77.71	62.06	77.47	90.87	82.87	82.03	60.05	65.27	53.59	64.07	46.62	69.49	20.5
GWD [30]	One	88.93	77.09	45.85	69.30	72.53	64.08	76.38	90.87	79.19	80.46	57.68	64.38	63.61	64.75	48.25	69.56	21.8
KLD [29]	One	89.51	79.98	39.94	70.38	78.04	64.24	82.85	90.90	81.78	83.04	57.63	63.52	56.63	65.16	50.05	70.24	21.6
KFIoU [47]	One	88.84	77.80	46.26	68.69	72.30	64.01	75.00	90.90	82.33	79.57	59.65	67.45	63.00	67.11	40.97	69.59	21.6
TIOE-R-ATSS [48]	One	88.72	76.55	51.52	70.25	77.84	75.48	82.32	90.88	77.36	85.81	58.41	69.65	64.93	66.89	45.76	72.16	22.0
NSD-T	One	89.20	81.86	49.98	72.03	76.55	73.09	82.77	90.91	86.15	85.69	62.98	68.58	65.45	70.94	69.93	75.07	-
NSD-S	One	88.89	80.18	46.99	70.95	76.24	72.09	82.44	90.80	86.16	84.89	61.79	63.05	64.04	67.56	60.44	73.10	31.5
NSD-S†	One	89.22	84.00	52.46	74.21	77.75	77.24	83.87	90.87	84.72	86.63	63.66	70.00	70.62	72.66	69.71	76.51	31.5

Note: "One", "Two", "Refine" represent the one-stage method, two-stage method and refine-stage method, respectively. Speed is the test result on single NVIDIA GeForce RTX 3090. The speed is computed by the tool in mmRotate framework. NSD-T means the teacher and NSD-S means the student after distillation, with the model architecture being R-ATSS with ResNet18.

a comprehensive experiment is set up in the final NSD framework, as shown in Table VIII.

In the NFA module, traditional positive samples improve student performance on three metrics, but there is still a gap

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TABLE XII
COMPARISON RESULTS OF THE DIFFERENT METHODS ON THE DIOR DATASET, WHERE THE VALUE IN BOLD IS THE BEST

Methods	Backbone	mAP	Speed (FPS)
R-RetinaNet [20]	ResNet-50	54.67	35.2
R-Faster RCNN [22]	ResNet-50	59.98	16.5
R-RepPoints [42]	ResNet-50	38.18	30.5
R^3Det [27]	ResNet-50	48.90	20.1
RoI-Trans [25]	ResNet-50	62.63	6.6
RoI-Trans [25]	Swin-T	65.29	15.0
R-ATSS [37]	ResNet-50	60.62	37.5
ReDet [28]	ReResNet-50	**66.46**	14.0
Oriented RCNN [40]	ResNet-50	62.42	17.1
KLD [29]	ResNet-50	55.60	37.2
KFIoU [47]	DarkNet-53	54.26	22.5
SASM [44]	ResNet-50	55.91	31.7
ARC-Oriented R-CNN [41]	ResNet-50	63.37	16.3
TIOE-R-ATSS [48]	ResNet-50	61.43	37.2
NSD-T	Swin-L	66.28	-
NSD-S	ResNet-18	**61.51**	**42.4**

compared to the features corresponding to our proposed NCS. In the NAP module, distilling traditional positive samples has almost no effect, which aligns with our previous demonstrations. In contrast, student performance exceeds that of single-feature distillation after distillation with the prediction outputs corresponding to our proposed NCS, proving the value of the detection head's prediction output as advanced semantic and localization knowledge. In addition, within the NAP module, the CD method tends to enhance mAP_{50}, whereas the LD method leans toward improving mAP_{75}, aligning, respectively, with the semantic and localization knowledge distilled. Ultimately, under the comprehensive NSD framework, the student still achieves higher performance after the distillation of the proposed NCS.

6) Evaluation of Efficiency: Employing a RetinaNet model with a ResNet backbone and a Swin Transformer, this study delineates the loss convergence of the student models during the training phase, with a comparative analysis of scenarios involving distillation and without distillation, as shown in Fig. 10. Notably, the NSD framework significantly accelerates the student model's convergence and yields a notably reduced loss.

We take the same model as described above for an example to evaluate the efficiency parameters of the models, including FLOPs, Params, and FPS, as shown in Table IX. Intriguingly, the student model retains identical efficiency metrics pre- and postdistillation. This consistency is attributed to the NSD framework's exclusive operation during training, devoid of any modifications to the model's parameters or its involvement in the inference stage. Furthermore, the lightweight student models achieve detection accuracy comparable to that of the teacher models, while exhibiting substantially lower parameter counts and FLOPs, and obtain significantly higher inference speeds than the teacher models. To provide a more intuitive reflection of speed and accuracy for the student and teacher models, we visualize the accuracy and speed of each model in Fig. 11. It is imperative to note that the result presented here is merely an illustrative example, as our approach exhibits good generalization capability and can be applied to models with larger disparities between the student and the teacher.

C. Distillation of More Detectors and Backbones

To validate the universality and effectiveness of our method, we conduct experiments on a variety of detectors and backbones. Our approach is applicable across heterogeneous backbones, such as distilling a ResNext-based teacher model into a ResNet-based student model, and even from a Swin Transformer-based teacher model to a ResNet-based student model. As shown in Table X, the outcomes reveal significant improvements in mAP and mAP_{75} metrics for student models distilled from various heterogeneous backbones teacher models. Moreover, student models perform better under the guidance of stronger teacher models. For instance, RetinaNet–ResNet18 achieves mAP scores of 69.9% and 70.4% after distillation from teacher models based on ResNet101 and Swin-L, respectively. Distillation from a Swin-L-based teacher also surpassed that from a Swin-B-based teacher.

In the experimentation with anchor-based two-stage detectors, a disparity in the RoIs between the teacher and student models is noted. To address these features from the teacher that correspond to the student's RoIs are fed into the teacher's head for prediction. Subsequently, the NAP module is employed to distill the predictive outcomes. This strategy ensures the effectiveness of our method and is easy to implement. Furthermore, we conduct distillation using RetinaNet based on Swin-T as the student model. To maintain the gap between the student and the teacher, the teacher model incorporates multiscale training.

D. Comparisons With State-of-the-Art Methods

In the comparative analysis with SOTA detectors and KD methods, we adopt a consistent approach by utilizing the classical R-ATSS as the detection framework, as done in our ablation studies. ResNet18 and ResNet101 are employed as the backbone networks for the student and teacher models and are referred to as NSD-S and NSD-T, respectively.

1) Results on DOTA: The DOTA dataset, a cornerstone in the realm of oriented object detection within remote sensing imagery, serves as a benchmark to evaluate various oriented object detection algorithms. We meticulously compare a suite of algorithms on the DOTA dataset. These algorithms include the classic R-RetinaNet, the newly proposed Oriented RepPoints, and so on, which are all implemented on the mmRotate framework. We maintain strict adherence to the experimental protocols of the mmRotate framework to ensure experimental fairness, utilizing the uniform ResNet50 as the backbone network across all models. The results are shown in Table XI, our lightweight model achieves a remarkable 72% mAP after postimplementation of NSD, which outperforms the baseline by 6% and eclipsing several existing methods, including advanced two-stage and refinement techniques. Notably, the incorporation of multiscale training in conjunction

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Fig. 13. Visualized results of detecting arbitrary-oriented aerial objects with our NSD on the DIOR dataset. The detected bounding boxes compactly enclose the objects.



Fig. 14. Visualized results of detecting arbitrary-oriented aerial objects with our NSD on the HRSC2016 dataset. The detected bounding boxes compactly enclose the objects.

with NSD further enhances the student model's performance, astonishingly elevating the mAP to 75%. It is important to underscore that NSD is confined to the training phase, leaving the network's inference speed unaffected. The partial detection results of the student model postdistillation on the DOTA dataset are displayed in Fig. 12. Our model exhibits superior performance in categories such as GTF and HL. These results indicate that NSD can assist lightweight models in effectively distinguishing objects and achieving accurate orientation predictions.

2) Results on DIOR: To further substantiate the efficacy of our proposed methodology, we extend our experiments to the extensive DIOR dataset. Specifically, we compare several classic and newly proposed oriented object detection algorithms, as shown in Table XII. Our lightweight model not only maintains an impressive image processing speed but also achieves a 70% mAP, which represents a 5% improvement over the baseline and narrows the performance gap with the teacher model to a mere 1%. The addition of multiscale training enables the student model to even surpass the teacher's performance. The visual depictions of the student model's postdistillation performance on the DIOR dataset are provided in Fig. 13.

3) Results on HRSC2016: To validate the effectiveness of the proposed NSD method in enhancing the localization performance of lightweight models, we compare the performance of various oriented detection algorithms on the HRSC2016 dataset, which contains a large number of remote sensing ships with high aspect ratios. As shown in Table XIII, our method achieves a mAP of 90.16%, the highest performance among the compared methods. Furthermore, it also reaches 60% in mAP_{75} and 55% in $mAP_{50:95}$ metrics, demonstrating excellent performance in localization. The visual results of the distilled student model on the HRSC2016 dataset are illustrated in Fig. 14.

TABLE XIII

COMPARISON RESULTS OF THE DIFFERENT METHODS ON THE HRSC2016 DATASET, WHERE THE VALUE IN BOLD IS THE BEST

Methods	mAP	mAP_{75}	$mAP_{50:95}$
R-RetinaNet [20]	72.50	36.80	41.39
R-Faster RCNN [22]	76.41	33.71	33.87
R-ATSS [37]	87.25	56.35	51.91
KLD [29]	84.11	27.90	37.60
Oriented RepPoints [45]	87.00	60.50	**55.79**
SASM [44]	87.40	56.10	50.95
ARC-Oriented R-CNN [41]	**88.27**	57.43	51.24
TIOE-R-ATSS [48]	87.58	57.88	52.95
NSD-T	88.55	64.48	54.80
NSD-S	**87.96**	**64.71**	**54.60**

E. Comparisons With State-of-the-Art KD Methods

In this study, we conduct a comprehensive evaluation of our proposed method against a range of SOTA KD methods on the DOTA dataset. To maintain experimental parity, the RetinaNet detector is uniformly employed as a baseline across all methods. We examine the distillation effects of various methods on two different backbones: ResNet and Swin Transformer. The detection performance on the DOTA dataset of models distilled through each method is reported in Table XIV. Notably, our method achieves mAP scores of 69.87% and 71% on the ResNet and Swin Transformer

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TABLE XIV
COMPARISON RESULTS WITH STATE-OF-THE-ART KD METHODS ON THE DOTA DATASET, WHERE THE VALUE IN BOLD IS THE BEST

	Methods	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
ResNet	Baseline	88.87	72.24	34.93	58.51	73.64	55.49	76.43	90.85	75.47	79.64	44.85	60.26	49.97	56.87	32.00	63.33
	DeFeat [6]	88.90	79.79	39.88	68.17	**77.21**	63.23	77.68	90.88	79.88	82.88	**59.58**	63.54	55.88	65.69	44.25	69.16
	FKD [50]	88.70	75.37	38.32	66.99	76.52	62.11	77.33	**90.90**	75.33	81.40	55.49	64.37	52.68	65.08	30.44	66.73
	LD [14]	89.31	78.07	38.77	67.88	75.56	60.31	77.17	90.89	80.92	81.15	58.22	62.01	54.35	64.63	42.15	68.09
	MGD [9]	88.77	76.02	**42.35**	67.30	74.87	65.08	78.05	90.89	77.73	**83.23**	57.44	59.57	56.66	64.95	40.68	68.24
	FGD [8]	88.72	72.21	41.07	65.72	76.75	**67.48**	**83.09**	90.87	77.79	81.54	55.07	62.69	55.93	**67.84**	44.74	68.77
	BCKD [51]	89.32	78.58	40.57	67.45	76.49	62.12	77.64	90.89	81.88	82.03	58.69	61.81	55.40	65.13	42.21	68.68
	OD [52]	**89.34**	78.30	39.41	70.65	76.48	63.23	77.68	90.84	82.94	82.63	56.61	**64.92**	55.61	67.24	44.07	69.33
	NSD(ours)	89.32	**82.05**	38.94	**71.29**	77.02	65.10	77.95	90.88	**83.06**	81.51	56.41	64.90	**57.23**	67.52	**44.81**	**69.87**
Swin Trans	Baseline	89.40	79.70	41.10	71.73	77.17	64.49	77.90	90.89	84.88	84.00	56.11	63.48	52.82	70.41	42.98	69.80
	DeFeat [6]	89.00	83.63	42.99	73.31	78.08	66.68	81.16	90.87	84.90	82.56	57.46	68.71	61.47	71.64	55.79	72.55
	FKD [50]	89.29	83.08	42.65	74.20	78.34	68.00	81.70	90.88	86.95	83.29	60.88	65.70	60.55	**72.48**	47.71	72.35
	LD [14]	89.57	82.66	38.74	72.97	76.52	65.09	78.20	90.87	85.91	83.56	61.66	64.16	57.15	69.57	53.80	71.36
	MGD [9]	89.24	83.17	43.43	73.41	78.47	68.27	83.50	90.90	86.29	83.69	61.09	**69.53**	62.40	71.62	57.62	73.51
	FGD [8]	89.22	83.72	**43.97**	74.49	**78.67**	69.43	**84.40**	90.77	**87.67**	83.49	60.78	69.46	**62.63**	71.18	57.91	73.85
	BCKD [51]	**89.61**	83.62	40.25	73.63	77.14	65.63	81.34	90.82	84.98	83.52	**61.81**	65.37	59.26	71.56	52.49	72.07
	OD [52]	89.56	81.22	43.12	75.47	76.48	68.23	81.45	90.88	84.89	83.76	61.17	68.36	57.92	70.85	58.25	72.77
	NSD(ours)	89.60	**84.66**	41.28	**76.75**	78.45	**70.01**	84.31	**90.90**	86.38	**84.51**	60.37	66.85	62.56	71.12	**60.84**	**73.90**

backbones, respectively, which are notably superior to those attained by several advanced distillation methods. A further enhancement is observed with the integration of multiscale training, where our method impressively records mAP scores of 75% and 79% on the respective backbones.

Category-Specific Performance: Postdistillation by our method, the models demonstrate superior performance in challenging categories with difficult-to-distinguish and densely arranged objects, such as baseball diamonds (BD), GTF, LV, and basketball courts (BC). This efficacy indicates our method's robustness in enhancing model performance across varied object scenarios.

Orientation Precision Analysis: In addition to mAP, we also evaluate our method using the mAoE metric, which is critical in oriented object detection tasks. Our method demonstrated the lowest mAoE among the compared techniques, indicating a significant enhancement in the models' precision for orientation prediction postdistillation. This result is particularly noteworthy considering the angular problems inherent in oriented object detection tasks.

Methodological Advantage: Our findings indicate that certain existing KD methods exhibit limitations, particularly when applied to the Swin Transformer backbone. In contrast, our approach effectively improves the performance of student models based on both ResNet and Swin Transformer backbones, highlighting the versatility and effectiveness of our proposed distillation method.

F. Analysis and Discussion

The efficacy of the NSD method has been corroborated across a variety of models and datasets. This is evidenced by the data presented in Figs. 12–14 and in Tables XI–XIII. These results collectively demonstrate the NSD framework's ability to consistently augment the performance of lightweight student models. Notably, as shown in Fig. 12(a) and (b), it is observed that conventional lightweight networks often face challenges in accurately identifying and distinguishing between densely clustered objects and intricate background patterns. In contrast, networks refined through the distillation process exhibit enhanced proficiency in discerning objects amidst complex backgrounds, thereby achieving more accurate object detection. This enhancement is primarily attributed to the NFA module, which contributes superior-quality features aiding in the semantic-level differentiation of objects from background elements. Moreover, the NAP module further bolsters the model's discernment capabilities by providing additional categorical supervisory signals during the distillation in the classification branch.

The NSD framework incorporates a novel distillation approach in the regression branch, specifically addressing the challenge of angle periodicity in both the prediction and distillation phases. This strategic implementation significantly enhances the model's localization accuracy. As shown in Fig. 12(c) and (d), the original network faces challenges such as misaligned bounding boxes and inaccurate orientation prediction for oriented objects. Yet, the distilled network achieves more accurate bounding box localization and orientation prediction.

V. CONCLUSION

Distilling oriented object detectors for remote sensing scenes has been a challenging task. Current mainstream distillation methods are designed for natural scenes and horizontal object detectors, which fail to effectively address the

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issues of foreground–background discrimination and inaccurate orientation prediction in lightweight detectors for remote sensing scenes. In this article, we design an NSD framework specifically tailored for distilling oriented object detectors in remote sensing images. NSD consists of two modules: negative-sample feature assimilation and negative-sample anti-false alarm prediction. The NFA module distills the features corresponding to negative samples from the neck to obtain discriminative and high-quality features. The NAP module distills the predictions corresponding to negative samples from the head and performs classification-based distillation to refine the orientation information of the objects. NSD has been extensively evaluated on multiple publicly available remote sensing datasets, and the distilled models consistently achieve superior performance. The effectiveness of the proposed method is validated through extensive experiments.

Based on the above research, we have turned our attention to the ViT [53] structure. In recent years, Transformer-based detectors have achieved SOTA accuracy, however, the substantial computational demands of ViT limit its application on edge devices. We have also conducted distillation work using the Swin Transformer as the backbone and achieved promising results, though this work remains within the same model pipeline and does not deeply explore the feature relationships at different scales within the Transformer. Building on our current work, we further attempt to identify the relevant knowledge between the teacher and student models by calculating feature attention. In addition, we have developed a prediction method that combines classification and regression for angle prediction in oriented detection, facilitating knowledge transfer.

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